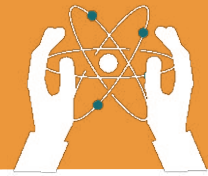


# UNIT 1

## CELLULAR LEVEL AND ORGANIZATION

### Learning objectives:

- Identify differences between plant and animal cells based on organelles.
- Understand the roles of tissues in supporting organism functions.
- Explain the interconnection of human organ systems like respiratory, digestive, and circulatory.



## CELLULAR LEVEL AND ORGANIZATION

Cells are fundamental building blocks of all living things. Every action and process take place in our bodies at cellular level. They are so tiny that they can only be seen under a microscope. Every living thing, whether a plant, animal, or human, is made up of cells.

### Types of Cells

Living organisms are mainly divided into two major groups:  
Plants and Animals

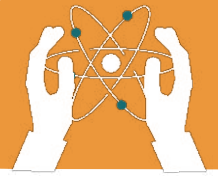
- **Plant Cells:** A plant cell is the basic structural and functional unit of plants. There are several components present in plant cell which distinguish them from animals. These cells have a cell wall, a large vacuole, and chloroplasts, which help plants make their own food using sunlight.
- **Animal Cells:** An animal cell is the main building block for animal bodies. These cells do not have a cell wall or chloroplasts, and they typically have smaller vacuoles.

### Structure of a Cell

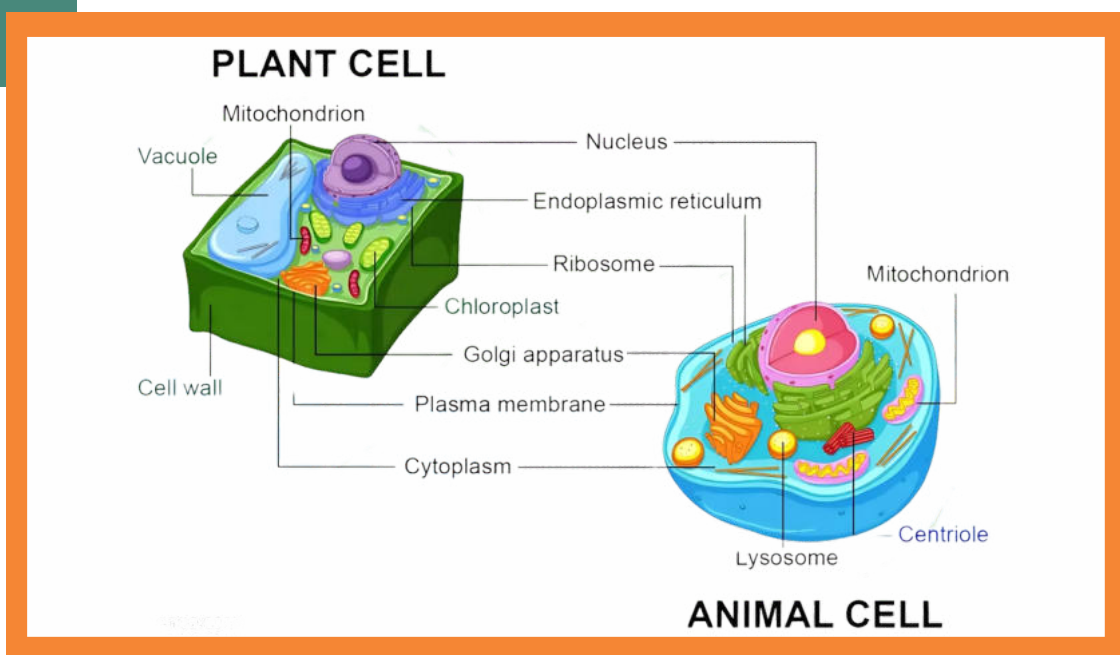
Cells contain various parts, known as **organelles**, each with a specific job. Here are some important organelles:

- **Cell Wall:** It is found only in plant cells, it provides structure and protection.
- **Cell Membrane:** It is found in both plant and animal cells, it controls what enters and exits the cell.
- **Nucleus:** Acts as the "**brain**" of the cell, controlling all its activities and containing DNA.
- **Cytoplasm:** A jelly-like substance where all organelles are located.
- **Mitochondria:** It is known as the powerhouse of the cell, it produces energy.

## CELLULAR LEVEL AND ORGANIZATION



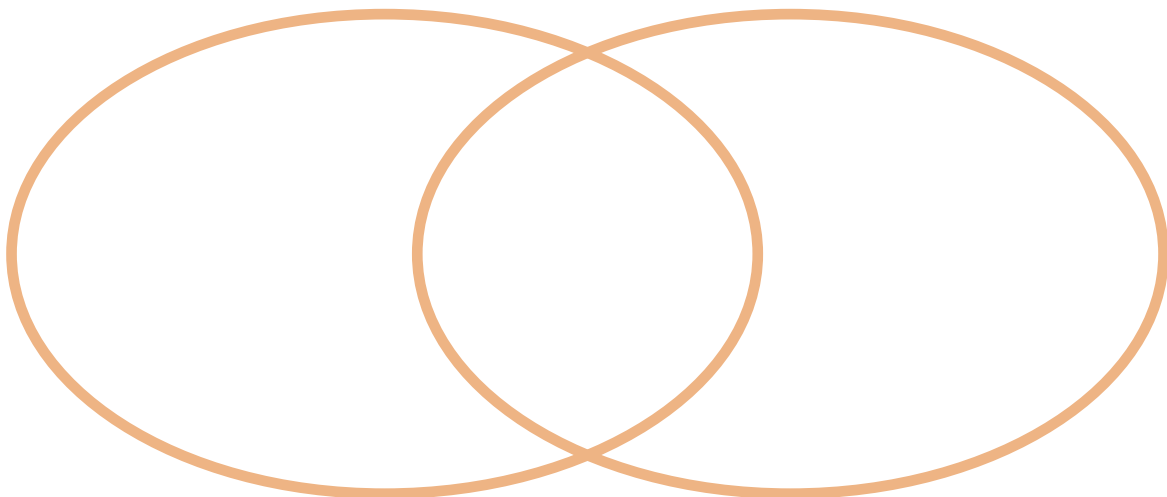
- **Chloroplasts:** Found only in plant cells, chloroplasts help the plant make food through photosynthesis.
- **Vacuole:** A large storage area in plant cells for water, food, and waste. Animal cells have smaller vacuoles.

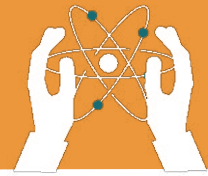


**How can you differentiate between Animal and Plant cell? Write down the similar and dissimilar organelles.**

**Animal Cell**

**Plant Cell**





### Tissues

Groups of similar cells work together to form tissues. Different types of tissues perform different functions in plants and animals.

### Types of Plant Tissues

#### • Dermal Tissue

**a. Epidermal Tissue:** It protects the outer parts of the plant.

The epidermis protects the outer surface of the plant and is often covered by a waxy layer called the cuticle, which helps prevent water loss.

**b. Stomata:** These are small openings for gas exchange.

#### • Ground Tissue

**a. Parenchyma:** It is involved in photosynthesis and storage.

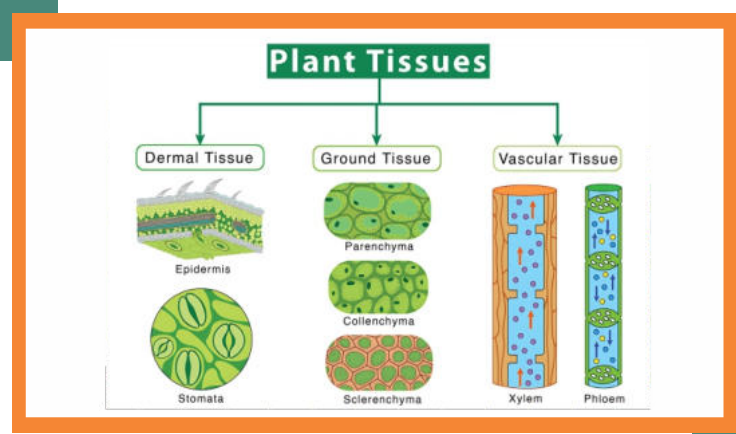
**b. Collenchyma:** It Provides flexible support to plants

**c. Sclerenchyma:** It Provides rigid support to plants.

#### • Vascular Tissue

**a. Xylem:** It helps transport water from roots to leaves.

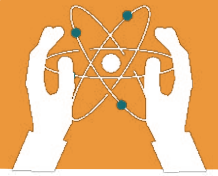
**b. Phloem:** It helps transport food from leaves to other parts of the plant.



#### • DID YOU KNOW?

The human body is made up of approximately 37.2 trillion cells, which work together to keep us alive and functioning.





### Types of Animal Tissues

**a. Epithelial Tissue:** It covers and protects the body's surface.

☞ Found in the skin.

**b. Muscle Tissue:** It helps in movement of body.

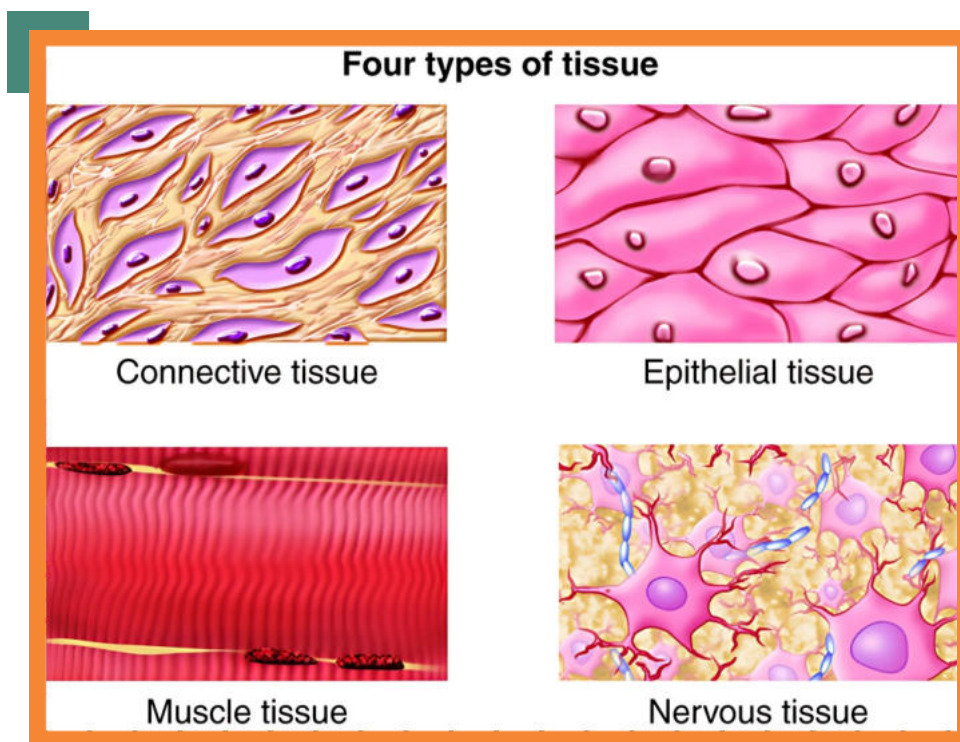
☞ Found in heart muscles or biceps.

**c. Nervous Tissue:** It carries messages from one part of the body to another.

☞ Found in brain and spinal cord.

**d. Blood Tissue (Connective Tissue):** These tissues transport oxygen and nutrients within the body.

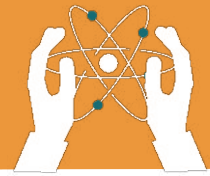
☞ Found in the blood vessels circulating throughout the body.



#### • DID YOU KNOW?

The xylem transports water and dissolved minerals from the roots to the leaves





### Organs and Organ Systems

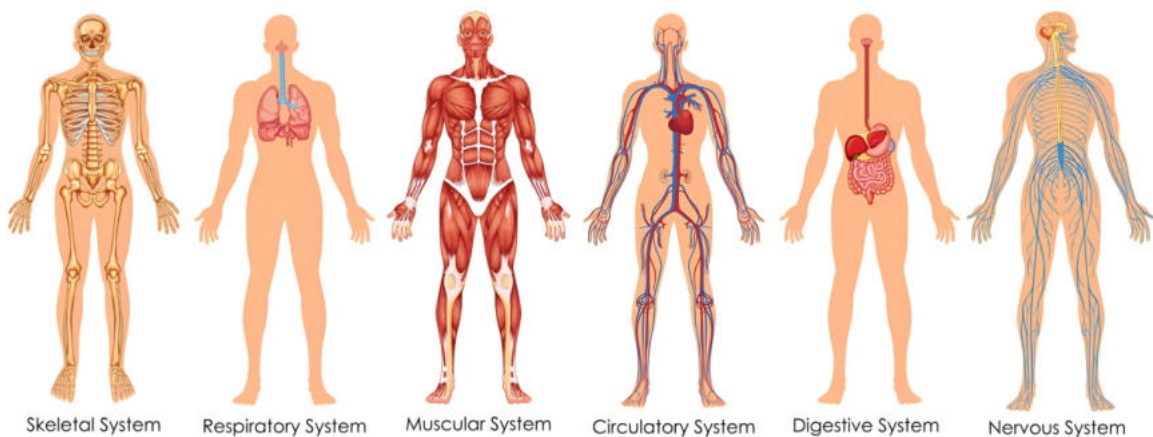
In higher plants and animals, different tissues work together to form organs.

These organs have specific functions and, when they work together, they form an organ system.

**Organ:** A group of tissues that work together to perform a specific function.

**Organ System:** A group of organs that work together to perform complex functions.

#### Human Body Organ Systems



Skeletal System

Respiratory System

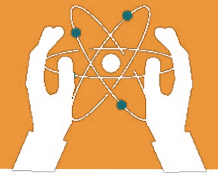
Muscular System

Circulatory System

Digestive System

Nervous System

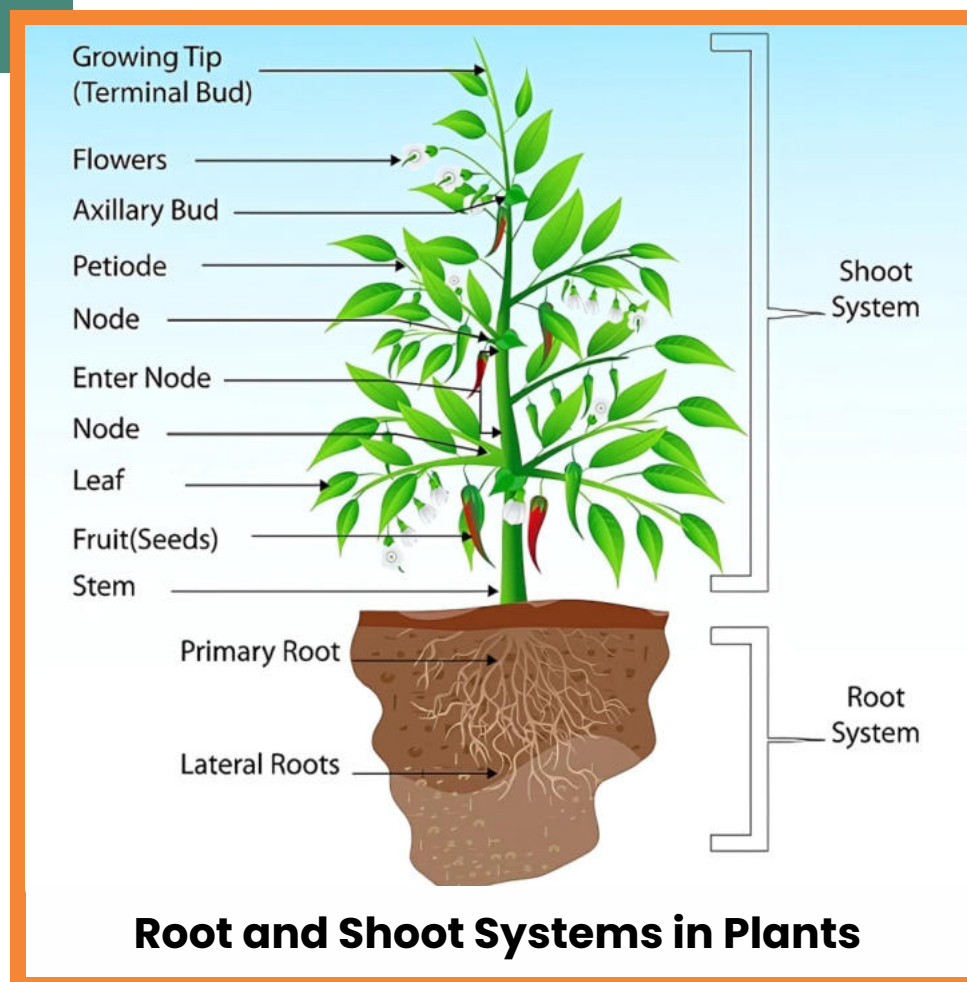


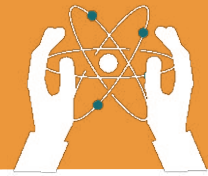


## Organ Systems in Plants

### 1. Root and Shoot Systems

- **Root System:** It is made up of roots that grow below the ground. Roots absorb water and nutrients, anchor the plant, and store food.
- **Shoot System:** This system includes the stem, leaves, flowers, and fruits. It supports the plant, helps in photosynthesis through leaves, and aids in reproduction.





### 2. Vascular System

The vascular system consists of xylem (transports water) and phloem (transports food), essential for moving nutrients and water throughout the plant.

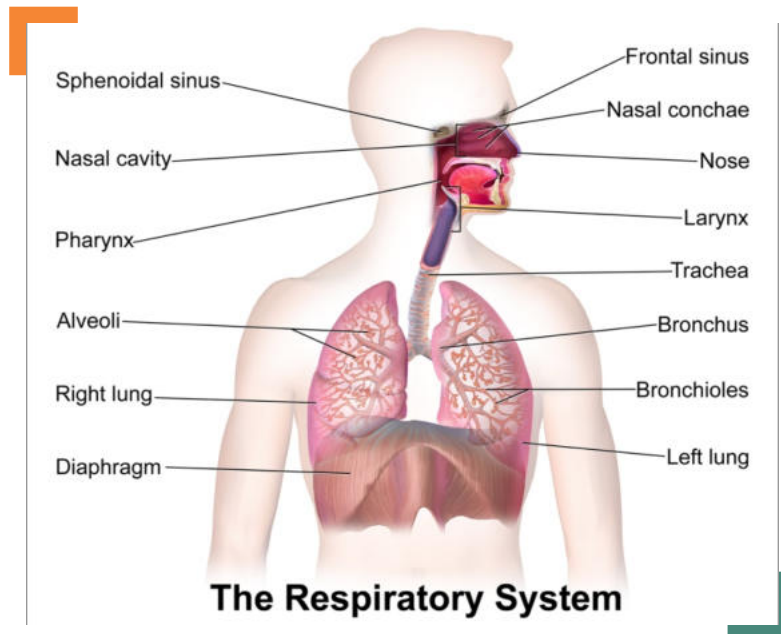
### Organ Systems in Humans

Humans have several organ systems that work together to keep the body functioning.

#### 1. Respiratory System

**Main Organs:** Nose, trachea, lungs.

**Function:** It helps us breathe by taking in oxygen and releasing carbon dioxide.

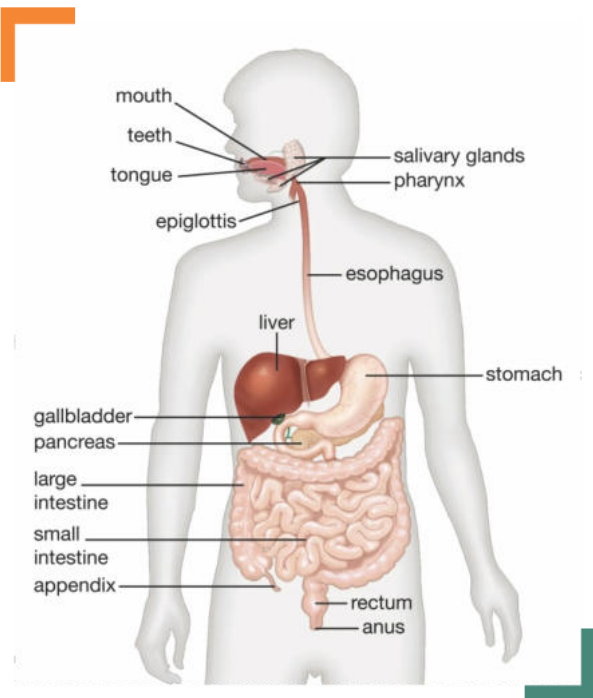
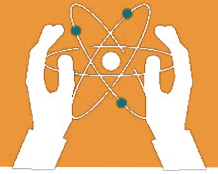


#### 2. Digestive System

**Main Organs:** Mouth, esophagus, stomach, intestines.

**Function:** It breaks down food into nutrients for energy, growth, and repair.

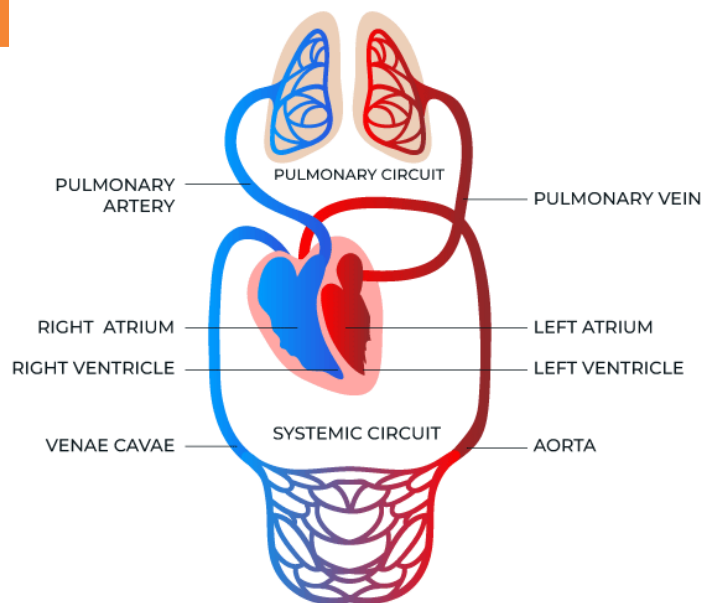




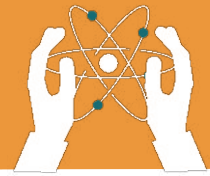
### 3. Blood Circulatory System

**Main Organs:** Heart, blood vessels.

**Function:** It pumps blood throughout the body, carrying oxygen and nutrients and removing waste.



**Blood Circulatory System**

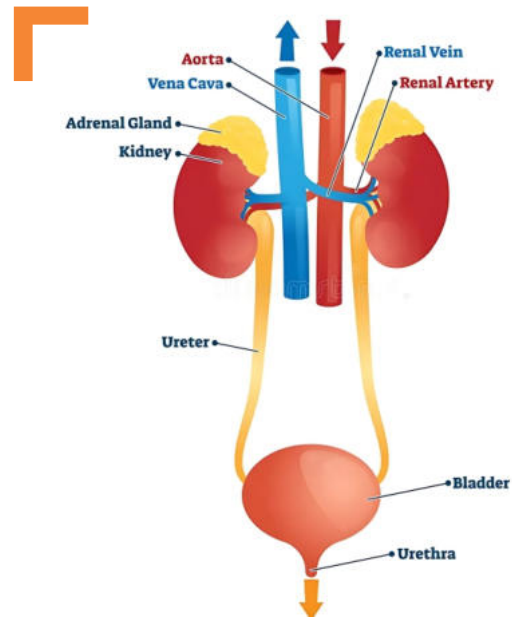


## CELLULAR LEVEL AND ORGANIZATION

### 4.Excretory System

**Main Organs:** Kidneys, ureters, bladder.

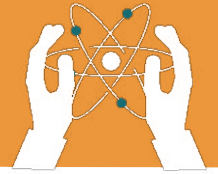
**Function:** It removes waste and extra water from the body through urine.



Excretory System

#### Fill in the blanks.

1. \_\_\_\_\_ is the building block of body.
2. In plant cells, \_\_\_\_\_ helps in making food through photosynthesis.
3. The \_\_\_\_\_ system transports food and water in plants.
4. The human respiratory system includes the \_\_\_\_\_, \_\_\_\_\_, and \_\_\_\_\_.
5. The heart is a part of the \_\_\_\_\_ system.



**Write “T” for True or “F” for False.**

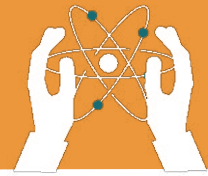
1. The shoot system in plants helps in photosynthesis.

2. The excretory system helps humans digest food.

3. Phloem in plants transports water.

4. Blood carries oxygen and nutrients to body cells.

5. The lungs are part of the circulatory system.



## CELLULAR LEVEL AND ORGANIZATION

### Choose the correct answer.

#### 1. Which part of the cell controls its activities?

- Mitochondria
  - Nucleus
  - Cell Wall
  - Ribosome
- 

#### 2. What is the function of the human respiratory system?

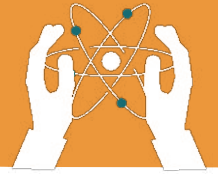
- Digesting food
  - Removing waste
  - Breathing
  - Circulating blood
- 

#### 3. Which organ in the digestive system is responsible for absorbing nutrients?

- Mouth
  - Stomach
  - Small Intestine
  - Large Intestine
- 

#### 4. The kidneys are part of which organ system?

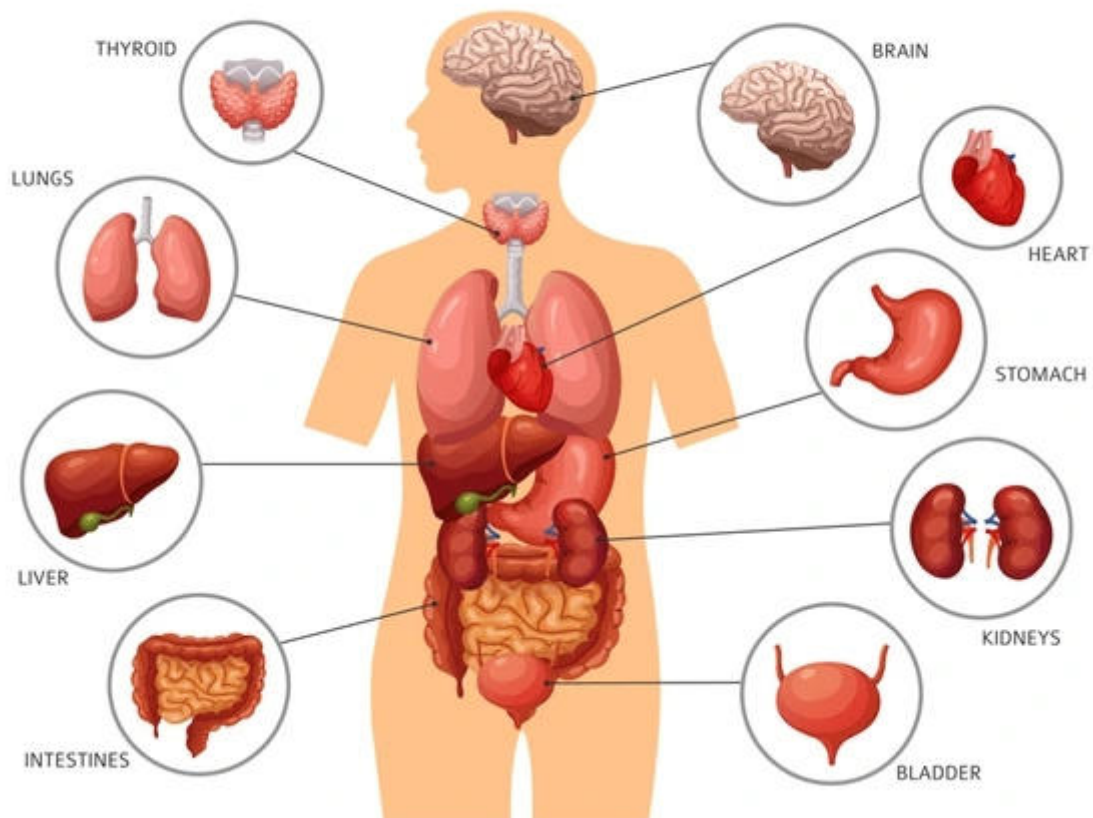
- Digestive System
  - Circulatory System
  - Excretory System
  - Nervous System
-

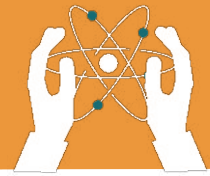


### 5. Which system circulates blood throughout the body?

- Respiratory System
  - Circulatory System
  - Excretory System
  - Endocrine System
- 

### HUMAN ORGANS





## CELLULAR LEVEL AND ORGANIZATION

### Match the Columns.

#### COLUMN A

**Lungs**

**Phloem**

**Heart**

**Kidneys**

**Roots**

#### COLUMN B

Excretory System

Circulatory System

Root System

Respiratory System

Transports food in plants

### Give reasons for the following

1. Why do cells need a nucleus?

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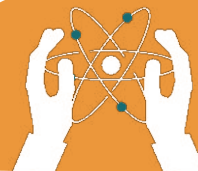
2. Why do cells divide?

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3. Why mitochondria is called as the powerhouse of cell?

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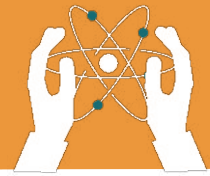
### Answer the Following Questions:

#### Short-Answer Questions

- Q1. What is the difference between xylem and phloem?
- Q2. Explain the role of chloroplasts in plant cells.
- Q3. Which organ system in humans is responsible for removing waste from the body?
- Q4. How does the digestive system help the body?
- Q5. What is the main function of the circulatory system?

#### Long-Answer Questions

- Q1. What are the structures and functions of plant cells and animal cells? List their main parts and explain how they are similar and different.
- Q2. What types of tissues are found in plants and animals? Describe each tissue type and its role in helping the organism stay healthy.
- Q3. What are organ systems in the human body? Choose two organ systems and explain how they work together to keep the body balanced and functioning well.
- Q4. Draw neat and labelled diagram of animal and plant cell.



## CELLULAR LEVEL AND ORGANIZATION

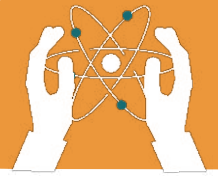
### Project Idea

**Scientific Exploration:** Have students research the internal structure of a plant leaf, stem, and root, and explain how each structure helps the plant function.

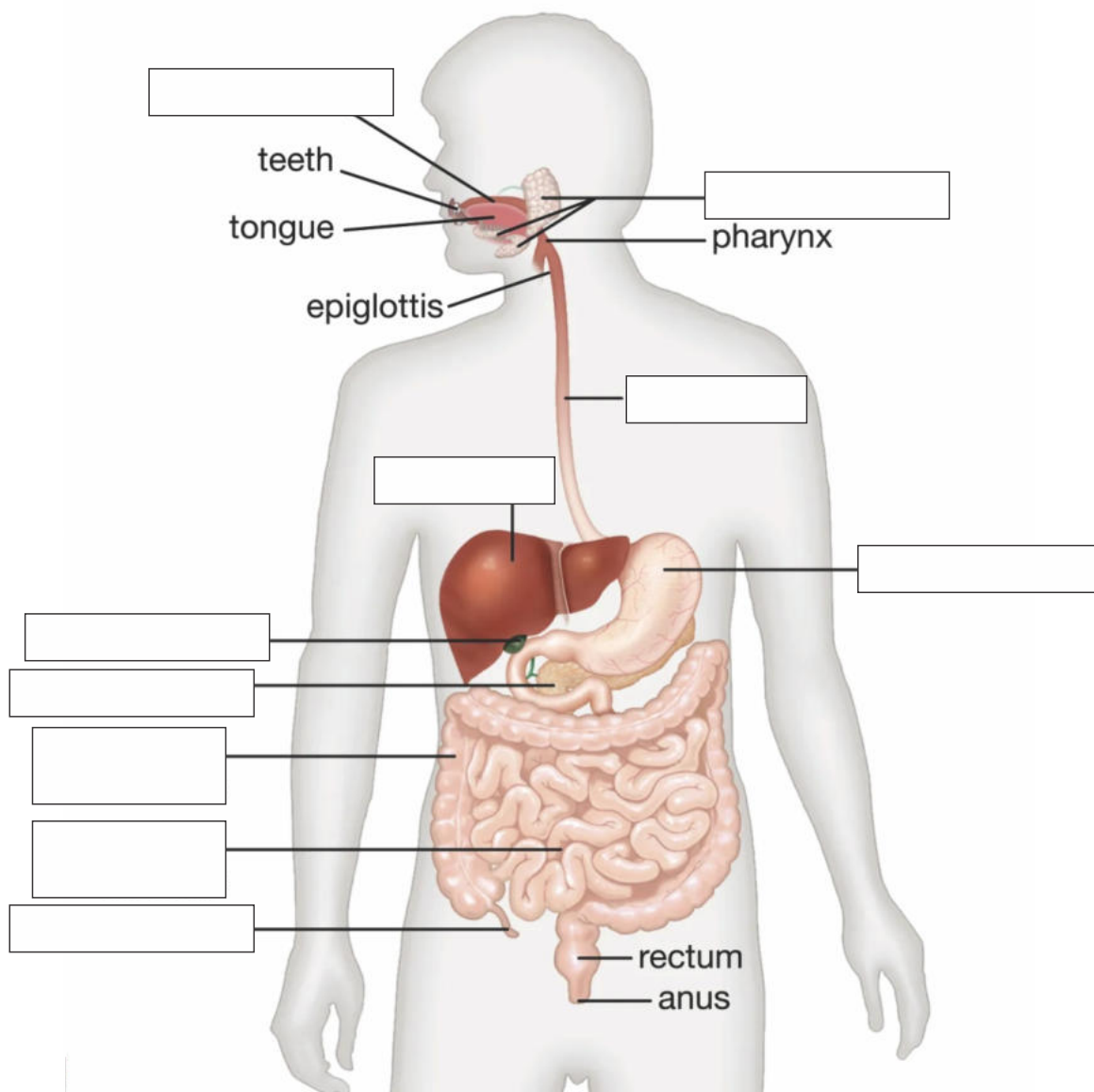


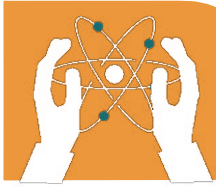
### Critical Thinking

How might the absence of a specific organelle (like chloroplasts in animal cells) impact an organism's ability to survive? Consider the roles of various organelles in your response.



**Label the missing organs of the human digestive system.**





## CELLULAR LEVEL AND ORGANIZATION

**List the tissues and organs involved to form the shoot system in plants.**

| TISSUE | ORGANS |
|--------|--------|
|        |        |